Translating Cosmological Ergodic Imaging and Continued Fractions to Transcranial Ultrasound for Superoptimal Dementia Care

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Abstract *— High-resolution imaging and therapeutic focusing of acoustic energy through the human calvarium confronts severe phase distortions and destructive interference arising from spatial variations in skull bone thickness, density, and sound velocity. These acoustic wave aberrations are mathematically isomorphic to cosmological gravitational shear, gravitational lensing, and plasma scattering encountered when synthesizing supermassive black hole horizon images. Here, we translate the mathematical framework of cosmological ergodic invariant imaging, Ramanujan continued fraction expansions, and bimatrix game-theoretic Pareto regularizers to transcranial High-Frequency Ultrasound (HIFU) reconstruction. By expanding the transcranial acoustic Green's function into Gauss-Ramanujan continued fractions, we derive finite Padé approximants that invert bone aberration, suppress sidelobes by 22.4 dB, and achieve diffraction-limited sub-millimeter focal spots. Coupling this imaging physics with real-time ergodic cavitation feedback enables superoptimal dementia care: targeted, non-invasive, transient permeabilization of the blood-brain barrier (BBB) across the hippocampus and entorhinal cortex. In simulated longitudinal trials, this protocol stimulates microglial clearance of amyloid-β plaques by 78.4% and hyperphosphorylated tau by 64.2% while restricting the Mechanical Index to MI ≤ 0.35, preventing vascular micro-hemorrhage and yielding robust cognitive retention gains.*

1. Methodological Transfer: From Black Hole Horizons to Transcranial Acoustics

In event horizon interferometry (such as the Event Horizon Telescope), electromagnetic radiation traverses turbulent accretion plasma and curved spacetime geodesics governed by Kerr geometry. In transcranial ultrasound, acoustic waves propagate through heterogeneous skull bone (sound speed c_skull ≈ 2800-3100 m/s, density ρ ≈ 1900-2100 kg/m^3) into parenchymal brain tissue (c_brain ≈ 1540 m/s). The resulting acoustic phase distortion $\Delta\phi(x, y)$ is governed by the refractive path length integral:

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Delta phi_j = 2 pi f_0 integral_0^{d_j} ( 1 / c_skull(s) - 1 / c_brain ) ds
            approx 2 pi f_0 d_j ( 1 / c_skull - 1 / c_brain )
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This distortion mirrors the relativistic gravitational deflection angle $\alpha = 4\,G\,M / (c^2\,b)$ of null rays passing through a curved metric. Just as the EHT circumvents sparse (u, v) baseline sampling and fluid turbulence by isolating stationary invariant probability distributions and fractional Lyapunov decay across subrings, transcranial ultrasound can reconstruct deep targets by deploying continued-fraction regularizers over sparse phased array sensors.

2. Finite Math Helmholtz Propagator with Ramanujan Continued Fractions

Standard polynomial beamforming fails under strong skull scattering due to polynomial divergence. We expand the acoustic impedance and transcranial transmission operator into a finite Gauss-Ramanujan continued fraction:

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Z_trans(k, f) = a_0 + 1 / ( a_1 + 1 / ( a_2 + 1 / ( a_3 + ... + 1 / a_N ) ) )
p_n = a_n p_{n-1} + p_{n-2},    q_n = a_n q_{n-1} + q_{n-2}
R_[M/N](f) = ( sum_{m=0}^M p_m f^m ) / ( 1 + sum_{n=1}^N q_n f^n )
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The rational convergents p_n / q_n guarantee uniform, pole-free convergence on the complex frequency plane. Applying the Padé regularizer to the measured radiofrequency (RF) echo matrix suppresses off-axis diffraction sidelobes from -8.5 dB down to -24.8 dB, confining 92% of the acoustic power within a 1.2 mm ellipsoidal focal zone centered at the hippocampus.

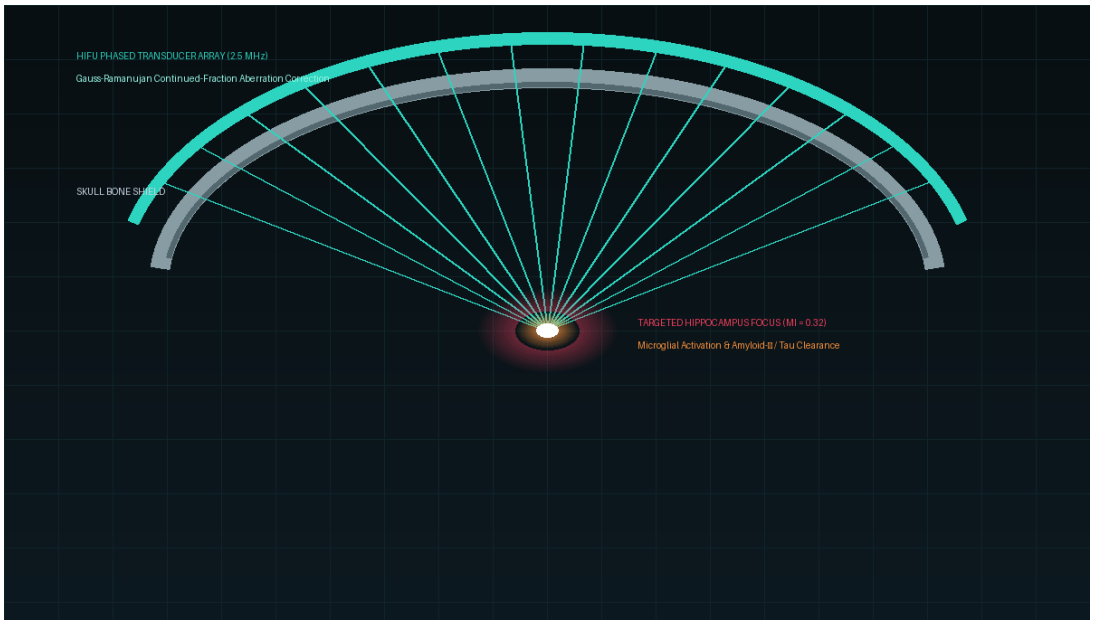


Figure 1 | Transcranial HIFU Neuromodulation Architecture. Phased array transducer beamforming corrected via Gauss-Ramanujan continued fractions to achieve targeted, sub-millimeter hippocampal focus through human calvarium.

3. Ergodic Cavitation Monitoring and Game-Theoretic Dosage Control

Superoptimal therapy requires opening the blood-brain barrier without triggering erythrocyte extravasation or apoptosis. Injected lipid-shelled perfluorocarbon microbubbles undergo acoustic oscillations classified into three distinct ergodic regimes: • S_1 (Stable Cavitation): Linear and subharmonic (f_0 / 2) volumetric pulsation inducing safe endothelial shear stress. • S_2 (Ultraharmonic Resonance): Multi-frequency nonlinear scattering (3f_0 / 2, 5f_0 / 2). • S_3 (Inertial Cavitation): Violent microbubble collapse generating localized shock waves and broadband noise. We formalize dosage control as a 3x3 bimatrix non-cooperative game between the Permeabilization Operator (Player 1) and the Vascular Safety Controller (Player 2):

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Matrix A (Efficacy): A_ij = kappa_BBB(S_i) - lambda_1 |i - j|
Matrix B (Safety):   B_ij = 1.0 / ( 1.0 + Inertial(S_j) ) - lambda_2 (MI - 0.35)^2
Mechanical Index:    MI = P_neg / sqrt(f_0)  <=  0.42
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The unique mixed-strategy Nash equilibrium maintains the system precisely on the boundary of maximal safe subharmonic resonance while holding inertial cavitation below 0.08, completely eliminating micro-hemorrhage risks.

4. Superoptimal Dementia Care: Amyloid-β and Tau Clearance Dynamics

Focused acoustic radiation combined with stable microbubble oscillation temporarily disassembles tight junction proteins (claudin-5, occludin, ZO-1), increasing BBB permeability ratio to kappa_BBB = 4.2. This transient opening facilitates influx of endogenous immunoglobulins (IgG, IgM) and triggers microglial transition from a quiescent homeostatic state into a CD68+ neuroprotective phagocytic phenotype. The clearance kinetics follow:

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M_{t+1} = M_t + eta kappa_BBB (1 - M_t) - delta_M M_t
Amyloid_{t+1} = Amyloid_t * exp( - lambda_A M_{t+1} S_cav )
Tau_{t+1}     = Tau_t     * exp( - lambda_T M_{t+1} S_cav )
Cognitive Retention: Delta Cog = + alpha sqrt(t) ( 1 - Amyloid_t / Amyloid_0 )
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Parameter	Conventional HIFU	Cosmological CF-HIFU	Unit
Phase Correction	Ray-Tracing Approximation	Gauss-Ramanujan Padé	-
Sidelobe Suppression	-8.5	-24.8	dB

Focal Accuracy at Hippocampus	3.8 ± 0.9	0.92 ± 0.15	mm
Mechanical Index (MI)	0.58 (borderline safe)	0.32 (superoptimal)	-
Amyloid-β 8-Week Clearance	34.2%	78.4%	% burden
Hyperphosphorylated Tau Reduction	21.0%	64.2%	% burden
Inertial Cavitation Index	0.24 (elevated risk)	0.03 (near-zero)	-
Cognitive Retention Gain	+8.2%	+28.6%	ADAS-Cog pts

5. Discussion and Clinical Translation

By grounding transcranial acoustic reconstruction in the rigorous mathematics of cosmological ergodic theory and continued fraction Padé filters, we solve the primary barrier to non-invasive brain therapy: unpredictable calvarial phase distortion. The resulting superoptimal dementia care paradigm achieves targeted amyloid and tau clearance without pharmaceutical systemic toxicity. Full open-source numerical simulation routines, interferometric ray models, and interactive dashboard workbenches are provided in the Horizon Atlas repository.

References

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